BBMS-13: Relational Database Schema and ER Diagram

Blood Bank Management System

Item	Details
Jira Task	BBMS-13 — Design relational database schema and ER diagram
Assigned Owner	Pavan E Naik
Database	Relational database / MySQL implementation
Deliverable	Database tables, schema, relationships, constraints and ER diagram

1. Identified Tables

The BBMS-13 task identifies the following entities/tables:

No.	Table / Entity	Purpose
1	User	Application user/account information
2	Role	User roles
3	Donor	Registered donor information
4	BloodGroup	Blood-group master data
5	Donation	Blood donation records
6	BloodUnit / Inventory	Blood stock and expiry information
7	BloodRequest	Blood request records
8	IssueRecord	Blood issue/distribution records
9	AuditLog	Audit records

2. Database-Relevant SRS Requirements

Req ID	Requirement	Database support
BBMS-F-003	Register donor with personal, contact and blood-group details.	Donor + User + BloodGroup
BBMS-F-004	Update donor personal/contact information.	Updateable Donor fields
BBMS-F-005	Staff view donor records.	Donor records
BBMS-F-006	Mark donor eligible / temporarily ineligible.	Donor eligibility status
BBMS-F-007	Record donor, blood group, quantity and donation date.	Donation table
BBMS-F-008	View donation history.	Donation linked to Donor
BBMS-F-009	Auto-update inventory after donation.	Donation → Inventory logic

BBMS-F-010	Maintain eight standard blood groups.	BloodGroup master
BBMS-F-011	View inventory summary by blood group.	Inventory + BloodGroup
BBMS-F-012	Search availability by blood group.	BloodGroup + Inventory
BBMS-F-013	Display available blood group and quantity.	Inventory
BBMS-F-014	Submit request with blood group, quantity and reason.	BloodRequest
BBMS-F-015	Track Pending/Approved/Rejected/Issued status.	BloodRequest.status
BBMS-F-016	Approve/reject pending request.	BloodRequest + User
BBMS-F-017	Issue blood against approved request.	IssueRecord + BloodRequest
BBMS-F-018	Reduce inventory when blood is issued.	IssueRecord → Inventory logic
BBMS-F-019	Record issue/distribution date, quantity and requester.	IssueRecord
BBMS-F-020	Create, update and deactivate user accounts.	User.account_status
BBMS-F-021	Filter/search donor, donation and request records.	Stored searchable fields
BBMS-F-022	Record expiry and exclude expired units.	Inventory.expiry_date

3.

Req ID	Requirement	Database/design implication
BBMS-SR-001	Protected pages require valid username/email and password.	User identity/authentication data required.
BBMS-SR-002	Administrative functions restricted to Administrator/Blood Bank Staff.	User → Role relationship.
BBMS-SR-004	Passwords must use secure hashing and never plaintext.	Store password hash, not plaintext.
BBMS-SR-005	Inventory/request-status changes must create audit entries.	AuditLog linked to User and affected record.
BBMS-NF-005	Inventory counts must remain consistent with donation and issue tran	sactions.

Security and Data-Integrity Requirements

Traceable Donation/IssueRecord transactions and inventory update log

Column	Data Type	Key / Constraint	Description
user_id	INT	PRIMARY KEY	Unique user ID
username_or_email	VARCHAR(255)	UNIQUE, NOT NULL	Login identity

4. Relational Schema

4.1 User

password_hash	VARCHAR(255)	NOT NULL	Secure hashed password
role_id	INT	FK → Role(role_id)	User role
account_status	VARCHAR(20)	NOT NULL	Active / Inactive

4.2 Role

Column	Data Type	Key / Constraint	Description
role_id	INT	PRIMARY KEY	Unique role ID
role_name	VARCHAR(50)	UNIQUE, NOT NULL	Administrator / Staff / Donor / Requester

4.3 Donor

Column	Data Type	Key / Constraint	Description
donor_id	INT	PRIMARY KEY	Unique donor ID
user_id	INT	FK → User(user_id)	Linked account
personal_details	TEXT	NOT NULL	Personal information required by SRS
contact_details	TEXT	NOT NULL	Contact information required by SRS
blood_group_id	INT	FK → BloodGroup	Donor blood group
eligibility_status	VARCHAR(30)	NOT NULL	Eligible / Temporarily Ineligible

4.4 BloodGroup

Column	Data Type	Key / Constraint	Description
blood_group_id	INT	PRIMARY KEY	Unique blood-group ID
blood_group	VARCHAR(3)	UNIQUE, NOT NULL	A+, A-, B+, B-, AB+, AB-, O+, O-

4. Relational Schema (continued)

4.5 Donation

Column	Data Type	Key / Constraint	Description
donation_id	INT	PRIMARY KEY	Unique donation ID
donor_id	INT	FK → Donor(donor_id)	Donor who donated
blood_group_id	INT	FK → BloodGroup	Blood group
quantity	INT	NOT NULL, > 0	Donation quantity
donation_date	DATE	NOT NULL	Date of donation

4.6 BloodUnit / Inventory

Column	Data Type	Key / Constraint	Description
inventory_id	INT	PRIMARY KEY	Inventory record ID
blood_group_id	INT	FK → BloodGroup	Blood group
quantity	INT	NOT NULL, ≥ 0	Current stock
collection_date	DATE	NOT NULL	Collection date
expiry_date	DATE	NOT NULL	Expiry date
Column	**Data Type**	**Key / Constraint**	**Description**
audit_id	INT	PRIMARY KEY	Unique audit ID
user_id	INT	FK → User(user_id)	User associated with action
action	VARCHAR(100)	NOT NULL	Action performed
entity_name	VARCHAR(100)	NOT NULL	Affected entity
entity_id	INT	NOT NULL	Affected record ID
created_at	DATETIME	NOT NULL	Audit time

4.7BloodRequest

Column	Data Type	Key / Constraint	Description
request_id	INT	PRIMARY KEY	Unique request ID
requester_user_id	INT	FK → User(user_id)	Requester
blood_group_id	INT	FK → BloodGroup	Requested blood group
quantity	INT	NOT NULL, > 0	Requested quantity
reason	TEXT	NOT NULL	Reason for request
status	VARCHAR(20)	NOT NULL	Pending / Approved / Rejected / Issued

4.8 IssueRecord

Column	Data Type	Key / Constraint	Description
issue_id	INT	PRIMARY KEY	Unique issue ID
request_id	INT	FK → BloodRequest	Related request
blood_group_id	INT	FK → BloodGroup	Issued blood group
quantity	INT	NOT NULL, > 0	Issued quantity
requester_user_id	INT	FK → User(user_id)	Requester
issued_by_user_id	INT	FK → User(user_id)	Staff member recording issue

issue_date	DATE	NOT NULL	Issue date

4.9 AuditLog

5. Relationships and Constraints

Parent	Child	Relationship	Foreign Key
Role	User	1 : Many	User.role_id → Role.role_id
User	Donor	1 : 0/1	Donor.user_id → User.user_id
BloodGroup	Donor	1 : Many	Donor.blood_group_id → BloodGroup.blood_group_id
Donor	Donation	1 : Many	Donation.donor_id → Donor.donor_id
BloodGroup	Donation	1 : Many	Donation.blood_group_id → BloodGroup.blood_group_id
BloodGroup	Inventory	1 : Many	Inventory.blood_group_id → BloodGroup.blood_group_id
User	BloodRequest	1 : Many	BloodRequest.requester_user_id → User.user_id
BloodGroup	BloodRequest	1 : Many	BloodRequest.blood_group_id → BloodGroup.blood_group_id
BloodRequest	IssueRecord	1 : Many	IssueRecord.request_id → BloodRequest.request_id
BloodGroup	IssueRecord	1 : Many	IssueRecord.blood_group_id → BloodGroup.blood_group_id
User	IssueRecord	1 : Many	IssueRecord.requester_user_id / issued_by_user_id → User.user_id
User	AuditLog	1 : Many	AuditLog.user_id → User.user_id

6. ER Diagram

Simple conceptual view of the Blood Bank Management System database

